

Homework 1(b) - Data Visualization

1 Bar Graph

The following data represents the number of students enrolled in different courses:

- Math: 120 students
- Physics: 80 students
- Chemistry: 100 students
- Biology: 90 students

1.1 Python Code for Bar Graph

```
import matplotlib.pyplot as plt

courses = ['Math', 'Physics', 'Chemistry', 'Biology'] students = [120, 80, 100, 90]

plt.bar(courses, students, color='blue', edgecolor='black') plt.xlabel('Courses') plt.ylabel('Students')
```

2 Pie Chart

A company recorded the sales distribution by product category:

- Electronics: 45%
- Furniture: 30%
- Clothing: 15%
- Other: 10%

2.1 Python Code for Pie Chart

```
import matplotlib.pyplot as plt

categories = ['Electronics', 'Furniture', 'Clothing', 'Other'] sales = [45, 30, 15, 10]

plt.pie(sales, labels=categories, autopct='%1.1f%%') plt.title('Sales Distribution by Product Category')
```

3 Pareto Chart

Given error types and their frequencies in a software application:

- Syntax: 50
- Logic: 30
- Runtime: 15
- Compilation: 5

3.1 Python Code for Pareto Chart

```
import matplotlib.pyplot as plt import numpy as np

error_types = ['Syntax', 'Logic', 'Runtime', 'Compilation'] frequencies = [50, 30, 15, 5]

sorted_freq = sorted(frequencies, reverse=True) cumulative_percentages = np.cumsum(sorted_freq)

fig, ax1 = plt.subplots() ax1.bar(error_types, sorted_freq, color='b') ax1.set_xlabel('Error Types')

ax2 = ax1.twinx() ax2.plot(error_types, cumulative_percentages, color='r', marker='o', label='Cumulative Percentages')

plt.title('Pareto Chart of Software Errors') plt.grid(True) plt.show()
```